

Assignment#03

1. Model Architecture

1.1 Design Overview

The embedding model follows a Siamese / shared-weight encoder architecture. A single backbone CNN processes each input image independently; the resulting feature vectors are L2-normalised to lie on the unit hypersphere before any loss is computed. This shared-weight design enforces that semantically similar images are mapped to nearby points in embedding space regardless of which branch they pass through.

1.2 Backbone

A lightweight convolutional backbone was used as the feature extractor. The final fully-connected classification head was replaced with a projection head consisting of:

- Global Average Pooling which collapses spatial dimensions
- Linear layer ($2048 \rightarrow 512$)
- Linear layer ($512 \rightarrow 128$) for final embedding dimension

1.3 Design Choices

The 128-dimensional embedding was chosen to balance expressiveness and retrieval speed. L2 normalization is critical for metric learning: it bounds all distances to $[0, 2]$ and makes the cosine similarity and Euclidean distance equivalent up to a constant factor, stabilizing training.

2. Dataset Sampling Strategy

2.1 Contrastive Pairs

For contrastive training, pairs of images are sampled as follows:

- Positive pairs (label = 0): two images drawn from the same class.
- Negative pairs (label = 1): two images drawn from different classes.

2.2 Random Triplet Sampling

Each triplet (anchor, positive, negative) is constructed by:

- Anchor: a randomly selected image from the dataset.
- Positive: a different image from the same class as the anchor.
- Negative: a randomly selected image from any other class.

Random sampling is simple and fast but frequently produces 'easy' triplets where the loss is already zero, leading to slow convergence. This is reflected in the low final Recall@1 of the random triplet model.

2.3 Hard Negative Mining

Hard mining selects the most informative triplets online (within each mini-batch):

- Hardest positive: the positive with the largest distance to the anchor (furthest same-class example).
- Hardest negative: the negative with the smallest distance to the anchor (closest different-class example).

3. Loss Functions

3.1 Contrastive Loss

The contrastive loss operates on pairs (x_i, x_j) with binary label $y = 0$ (same class) or $y = 1$ (different class) and a margin $m > 0$:

$$L_{\text{contrastive}} = (1 - y) \cdot d^2 + y \cdot \max(0, m - d)^2$$

where $d = \|f(x_i) - f(x_j)\|_2$ is the Euclidean distance between embeddings. When $y = 0$ (similar pair) the loss penalizes large distances; when $y = 1$ (dissimilar pair) the loss penalizes distances smaller than the margin m . Implementation note: margin m was set to 1.0 (consistent with unit-sphere embeddings). Pairs were sampled with equal class-balance per batch.

3.2 Triplet Loss

The triplet loss (Schroff et al., 2015) operates on triplets (a, p, n) anchor, positive, negative and enforces $d(a, p) + \alpha < d(a, n)$:

$$L_{\text{triplet}} = \max(0, d(a, p)^2 - d(a, n)^2 + \alpha)$$

where $\alpha > 0$ is the margin hyperparameter (set to 0.2 in this study). The loss is zero when the negative is already further than the positive by at least α . Implementation note: PyTorch's `nn.TripletMarginLoss` was used with $p = 2$. The same backbone and projection head are shared across anchor, positive, and negative branches.

3.3 Key Differences

Contrastive loss uses absolute distance thresholds and requires careful margin tuning. Triplet loss enforces a relative ordering (positive closer than negative) which is more naturally aligned with retrieval. Hard mining multiplies the effective learning signal by ensuring every triplet carries non-zero gradient.

4.Retrieval Performance:

4.1 Results Table

The table below summarizes final-epoch performance across all three methods. Recall@1 and Recall@5 is the fraction of queries for which the correct class appears in the top-k retrieved images.

Method	Recall@1	Recall@5	Final Loss	Epochs
Contrastive	0.6421	0.8593	0.1028	10
Triplet (Random)	0.6166	0.8156	0.0064	6
Triplet (Hard Mining)	0.7201	0.8426	0.1873	10

4.2 Analysis

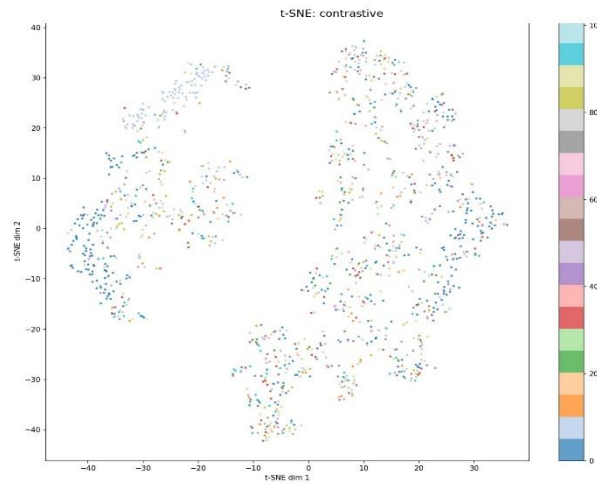
Hard negative mining delivers the strongest retrieval quality: Recall@1 of 0.7201 vs. 0.6421 (contrastive) and 0.6166 (random triplet), a relative improvement of +12% over contrastive and +17% over random triplet. Recall@5 follows the same ordering, confirming the ranking is well-calibrated at the top.

The random triplet model achieves the lowest Recall@1 despite converging to the lowest absolute training loss (0.0064 at epoch 6 before early stopping). This is a classic sign of trivial-triplet collapse: the model easily satisfies most randomly-sampled triplets but learns insufficient discriminative structure.

The contrastive model converges steadily over 10 epochs with monotonically decreasing loss (0.1615 to 0.1028), achieving a competitive Recall@5 of 0.8593 — the highest among the three models — indicating good coarse-level structure in the embedding space even if fine-grained nearest-neighbors precision lags behind hard mining.

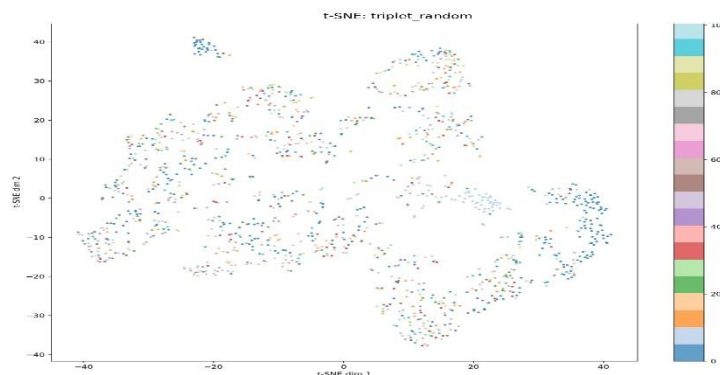
5.t_SNE Visualizations

t-SNE projects the learned 128-dimensional embeddings into 2D for qualitative inspection.



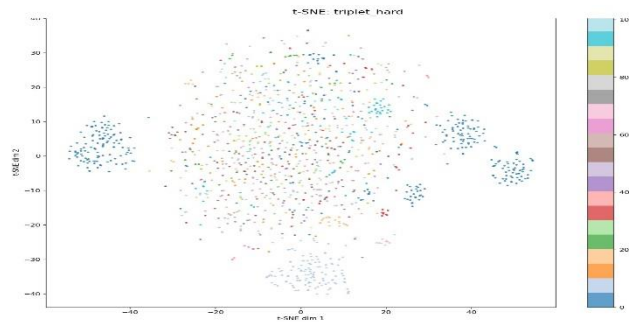
The contrastive embedding space forms an elongated, loosely structured manifold. There is a visible left–right separation with several coherent local clusters, particularly along the periphery, but the interior of the projection is densely mixed.

5.2 Triplet Loss Random Sampling Embeddings



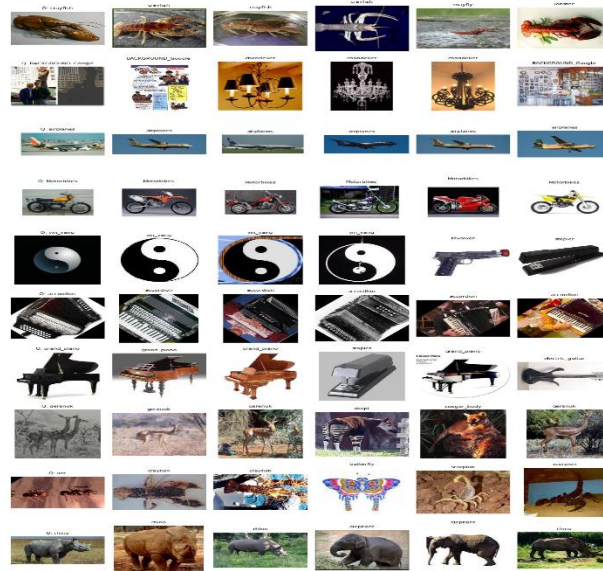
The random triplet embedding space resembles the contrastive plot in overall density and structure, with a few small peripheral clusters (top-left and top-right) but substantial mixing in the centre. The lack of clearly separated class regions is consistent with the quantitative result: random triplets do not force the model to resolve ambiguous class boundaries, leaving many classes collapsed in the same embedding region.

5.3 Triplet Loss Hard Mining Embeddings

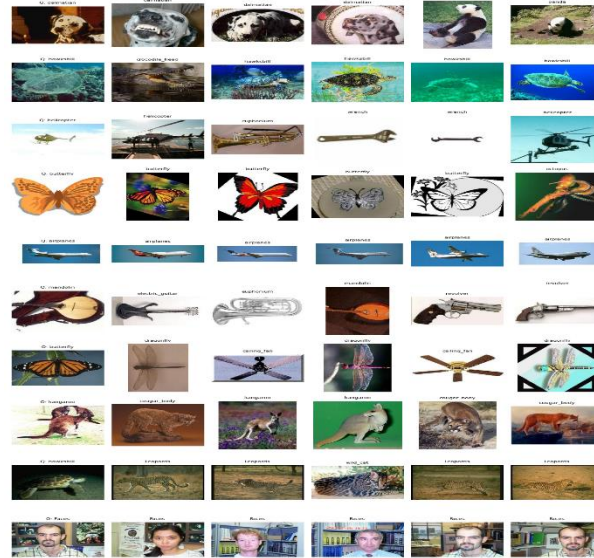


Hard mining produces the most structured embedding space of the three. Several classes form clearly separated satellite clusters (visible at the left, right, and bottom periphery), while the dense central mass still shows mixing.

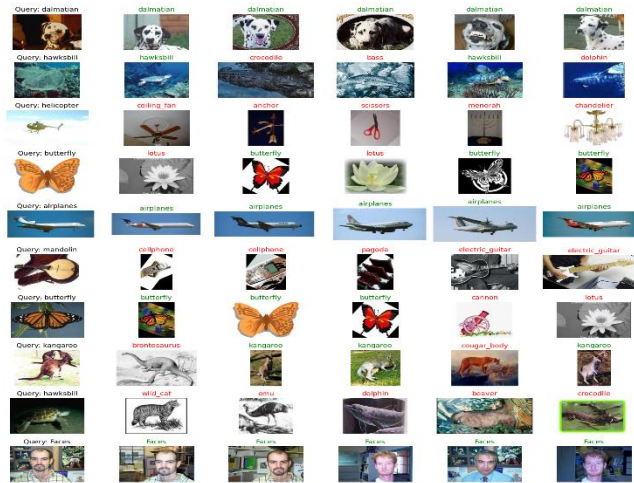
6. Retrieval Visualizations



Contrastive model: Retrieval is generally correct for the top-1 result but degrades at ranks 3–5. The model occasionally retrieves images from visually similar but semantically distinct classes, indicating that the margin-based separation is less precise than triplet hard mining.



Random triplet model: The highest error rate. For ambiguous queries, several retrieved images belong to incorrect classes even at rank 1. The model appears to cluster by low-level texture or colour rather than semantic content for difficult queries.



Hard mining model: For visually distinctive classes (e.g., aircraft, certain animal species), all 5 retrieved images are correct. Errors tend to occur between visually similar sub-categories (e.g., sedans vs. SUVs, various bird species with similar plumage).

7. Discussion

7.1 Comparison of Methods

Hard negative mining emerges as the clear winner in terms of Recall@1, confirming established findings in the metric learning literature. By focusing the loss on boundary-violating triplets, the model is compelled to resolve the most challenging class distinctions in every update. The cost is a higher and more volatile training loss, reflecting that the selected triplets are always non-trivial.

The contrastive model achieves the best Recall@5, suggesting it learns a well-ordered ranking at a broader scale even if its nearest-neighbour precision is lower. This can be explained by the absolute-distance objective: contrastive loss explicitly repels dissimilar pairs beyond a margin, which is beneficial for retrieving any correct item within the top 5 even if the top-1 is wrong.

Random triplet sampling is the weakest approach. Despite producing the lowest final training loss, the model collapses on trivial triplets and fails to learn fine-grained class structure. This underscores that loss magnitude alone is not a reliable indicator of embedding quality.

7.2 Failure Cases

Common failure modes across all methods include fine-grained sub-category confusion, background-dominated representations, and class imbalance. These could be addressed through:

- Curriculum-based mining: gradually increasing difficulty rather than always using the hardest negatives.
- Data augmentation: stronger augmentation to reduce background sensitivity.
- Proxy-based losses : avoid pair/triplet sampling entirely by learning class proxies.

7.3 Conclusions

This study demonstrates that the triplet loss with online hard negative mining outperforms both contrastive loss and random triplet sampling for image retrieval on a 100-class benchmark. The key insight is that mining quality that not loss type the primary driver of embedding quality. Future work should explore semi-hard mining (negatives inside the margin but outside the positive boundary), which can provide a better bias-variance trade-off than fully-hard mining while avoiding the trivial-triplet problem of random sampling.

CHATGPT LINK

<https://chatgpt.com/c/69c7d33d-bd24-83e8-b073-fd1f4fbf636f>

GITHUB LINK

https://github.com/bilalbushra789/Deep_Learning_Assignment.git

GOOGLE DRIVE LINK

<https://drive.google.com/drive/folders/1ft6U7vFrjZSUWIBKlvFqtGUh3UQnACpt?usp=sharing>